SS-MCNet: A Spatial–Spectral Multidimensional Collaborative Network for Fine-Grained Wetland Classification With UAV Hyperspectral Imagery

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***Abstract*—Wetlands are critical ecosystems where fine-grained land-cover classification is crucial for effective management. Although unmanned aerial vehicle (UAV)-based hyperspectral imaging provides both high spectral and spatial resolution, we observed significant confusion among spectrally and spatially similar classes in our self-collected Yellow River wetland dataset. To address this, we propose the spectral–spatial multidimensional collaborative network (SS-MCNet), which comprises a spectral probe module (SPM) for fine spectral representation, a spatial texture module (STM) for detailed texture extraction, and a spectral–spatial interaction module (SSIM) to model cross-domain dependencies. A collaborative attention fusion mechanism is introduced for adaptive balancing of spectral and spatial information, and a consistency loss (CL)–diversity loss (DL) function is designed to maintain consistency and complementarity among the three modules. Experimental results show that SS-MCNet achieves state-of-the-art performance with overall accuracy (OA) of 98.08%, average accuracy (AA) of 97.42%, and a kappa coefficient of 97.75% and performs well in an additional hyperspectral–panchromatic fusion experiment, providing a reference for multisource fusion classification.**

***Index Terms*—Deep learning, hyperspectral imaging, land-cover classification, unmanned aerial vehicle (UAV) remote sensing, Yellow River wetland.**

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I. Introduction

REMOTE sensing is critical for ecological monitoring, and wetland land-cover classification plays a key role in environmental protection and resource management [1], [2], [3], [4], [5], [6]. Compared with the multispectral imaging, hyperspectral sensing offers higher spectral resolution, enabling subtle land-cover differentiation and supporting species-level and complex object recognition [7]. Satellite-based hyperspectral systems suffer from limited spatial resolution and temporal constraints [8]. In contrast, unmanned aerial vehicle (UAV)-based hyperspectral remote sensing provides higher spatial detail, flexible acquisition, and improved imaging quality, offering a promising solution for fine-grained wetland classification [9], [10], [11], [12]. By combining rich spectral and fine spatial information, such data facilitates fine-grained classification of diverse land-cover types [13].

Analysis based on the self-constructed UAV hyperspectral dataset of the Yellow River wetland reveals that land-cover types with similar spectral bands and spatial textures are prone to significant classification confusion [14], [15], [16], [17]. To address this issue, it is necessary to build a feature modeling framework that accurately captures spectral bands (e.g., sequences of absorption peaks) and spatial textures (e.g., object contours) [18] to avoid feature overlap in a single domain [19], [20]. Meanwhile, a feature fusion strategy with dynamic cross-dimensional interaction capability should be designed to enhance the complementarity between spectral and spatial features, such as guiding texture selection with spectral cues or suppressing spectral noise using spatial constraints [21]. Breakthroughs in both feature modeling and fusion are essential to construct a joint feature representation with strong discriminative power to effectively alleviate classification confusion [22]. Various feature modeling and fusion methods have been proposed to improve the classification performance [23]. In hyperspectral feature modeling, early methods used traditional machine learning such as support vector machines (SVMs) [24] and random forests, relying on handcrafted features and suffering from the "curse of dimensionality." They lacked spatial information use and underperformed in

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complex scenes [25]. Recently, these methods remain practical in high-resolution, multisource remote sensing. For example, one study used feature selection to improve UAV multispectral mangrove classification, achieving high accuracy [26]; another combined multiscale variable selection with SAR time series for precise mangrove identification and scattering analysis [27]. These show traditional machine learning's robustness in feature modeling and multisource fusion. With deep learning, convolutional neural network (CNN) [28] greatly improved classification via automatic feature extraction. The 1-D-CNN captures local spectral changes; 2-D-CNN extracts spatial textures but treats spectral dimensions as channels, limiting spectral and spectral–spatial modeling [29]. The 3-D-CNN [30] uses volumetric convolutions to jointly process spectral and spatial data, improving over 2-D-CNN [31], but focuses on local voxels and misses global spatial structures [32], [33].To address this, the spectral–spatial residual network (SSRN) [34] was introduced to enhance global context modeling. Yang et al.'s [35] cross-mixing residual network (CMR-CNN) connects 3-D spectral and 2-D spatial features via AFE for fusion but relies on static stacking, limiting deep semantic modeling. Paoletti et al.'s [36] deep pyramid residual network (DPRnet) uses pyramid bottlenecks and multiscale downsampling for hierarchical features, addressing static fusion limits but risks detail loss from fixed scales. Cao et al.'s [37] mixed residual convolutions with Vision Transformer (MCVT) combines 2-D/3-D convolutions and ViT to capture global dependencies via self-attention, improving modeling but applying uniform coarse fusion, weakening fine spec tral and spatial detail. In hyperspectral feature fusion, early methods used simple concatenation [38]. Lin et al.'s [39] linear transformation concatenation fusion (LTCF) aligns features via linear mapping with rectified linear unit (ReLU) and dropout, but causes redundancy by stacking spectral and spatial features. Zhu et al.'s [40] adaptive-weighted fusion (AWF) uses SoftMax weights from dual 1×1 convolutions to highlight key features, yet remains static. Wang et al.'s [41] decision-level-weighted fusion (DWF) adjusts weights postclassification, adding flexibility but lacking deep fusion. Xie et al.'s [42] fully connected concatenation fusion (FCCF) uses ReLU and fully connected layers for dimension alignment but is computationally heavy and less adaptable. Li et al.'s [43] feature fusion gate framework (FFGF) learns nonlinear dependencies via gates for dynamic fusion, improving interaction but still relying on fully connected layers, limiting complex relationship modeling.

Based on a self-collected UAV hyperspectral dataset of the Yellow River wetlands, this article proposes a spatial–spectral multidimensional collaborative network (SS-MCNet), comprising the spatial–spectral multidimensional collaborative module group (SS-MCMG) and the spatial–spectral multidimensional collaborative attention fusion (SS-MCAF). SS-MCMG includes three complementary modules: spectral probe module (SPM), spatial texture module (STM), and spatial–spectral interaction module (SSIM), which independently and in parallel model spectral, spatial, and joint features. These modules integrate multiscale perception and position-aware sequence modeling to balance local detail extraction and global context understanding. SS-MCAF employs a three-round mutual attention mechanism, where the feature substreams from the three modules, under the principle of equal importance, dynamically and in parallel assume the roles of query (Q), key (K), and value (V), enabling deep feature collaboration and interaction. This design highlights the equivalence and complementarity of heterogeneous features, enhancing information exchange and fusion depth. To further stabilize fusion and prevent dominance by any single module, two loss functions are introduced: a consistency loss (CL) that constrains distributional consistency under different query configurations to improve training stability, and a diversity loss (DL) that promotes feature decorrelation and discriminability by minimizing correlation or maximizing mutual information. Together, these mechanisms enhance the model's stability, robustness, and accuracy in wetland hyperspectral classification.

Hyperspectral data excels in classification with rich spectral detail but suffers from low spatial resolution. Panchromatic images offer higher spatial clarity yet lack spectral distinction. Data fusion combines their strengths and has become vital for improving classification [44]. This study fuses panchromatic and hyperspectral imagery to build a high-resolution dataset, applying the proposed model for classification. Results show notable accuracy gains with few samples, validating the model's cross-source adaptability and fusion capability. Though not the study's core, this provides new insight into efficient multisource remote sensing classification.

The main contributions of this work are summarized as follows.

1) We propose SS-MCNet, integrating feature modeling and fusion via SS-MCMG and SS-MCAF, overcoming traditional separate processing limitations, significantly improving classification accuracy in complex scenarios.
2) SS-MCMG incorporates SPM, STM, and SSIM, combining multiscale perception and positional encoding to deeply model hyperspectral heterogeneous features from spectral, spatial, and cross-dimensional perspectives, providing high-quality feature representation.
3) SS-MCAF employs a three-round mutual attention mechanism for deep fusion of feature streams and introduces consistency and DLs to balance feature stability and complementarity. Comparative experiments validate its superiority over conventional strategies with marked classification performance gains.
4) This study not only focuses on a single classification model but also covers the entire workflow from UAV flight planning, hyperspectral data acquisition and processing, field survey and dataset construction, to analysis of hyperspectral multisource fusion data, demonstrating high systematization and practical applicability.

II. Proposed SS-MCNet

The proposed SS-MCNet framework, as illustrated in Fig. 1, consists of the processing flow from input hyperspectral images through SS-MCMG, followed by feature fusion via the SS-MCAF, then loss function computation, and finally producing the classification output.

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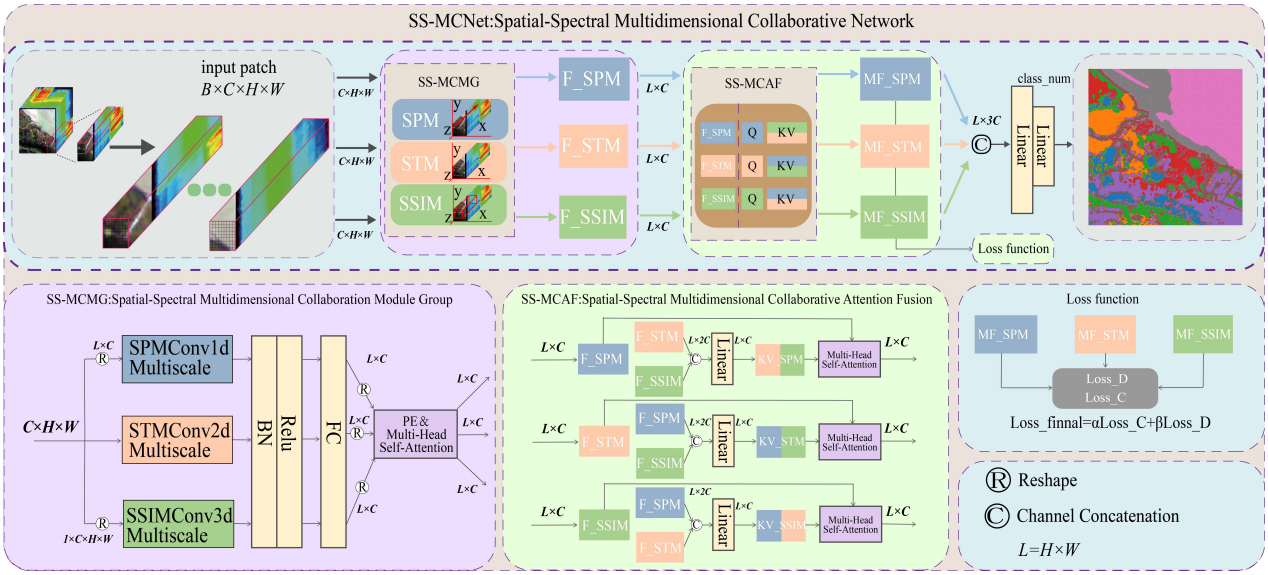


Fig. 1. Input patch (with dimension $B \times C \times H \times W$) is fed into SS-MCMG (containing SPM, STM, and SSIM submodules) to extract spectral, spatial, and spatial–spectral interaction features (F_SPM/F_STM/F_SSIM). Then, these features are fused via SS-MCAF. After multiscale processing (MF_SPM/MF_STM/MF_SSIM), channel concatenation is performed. Combined with a linear layer and a loss function, the classification result is output.

A. Spatial–Spectral Multidimensional Collaborative Module Group

SS-MCNet takes hyperspectral data $X \in \mathbb{R}^{B\times C\times H\times W}$ as input, where B is the batch size and C is the number of spectral bands, and H and W are the spatial resolutions. The input is fed in parallel into the SPM, STM, and SSIM. Each module employs a multiscale parallel convolution mechanism for feature extraction, utilizing convolutional kernels with different receptive field sizes to enhance adaptability to both local and global structures. The features are then projected uniformly to the Transformer embedding dimension C_1, and sinusoidal positional encoding is added along the sequence length $L = H \times W$. Finally, multihead self-attention models the global dependencies. The overall process can be expressed as follows:

$$F_{\text{SPM}}, F_{\text{STM}}, F_{\text{SSIM}} = \text{Transformer}_m\left(S_m + \text{PE}\left(L, C_1\right)\right) \quad (1)$$

$$S_m = \text{reshape}\left(\text{Conv}_m^{\text{MS}}\left(X\right), \left(B, L, C_1\right)\right)$$
$$m \in \{1-\text{D}, 2-\text{D}, 3-\text{D}\} \quad (2)$$

where $\text{Conv}_m^{\text{MS}}$ denotes the multiscale parallel convolution operation with output shape (B, C_1, H, W), $\text{PE}(L, C_1)$ is the sinusoidal positional encoding matrix, and $\text{Transformer}_m(\cdot)$ represents the multihead self-attention encoder.

To efficiently inject positional information without adding learnable parameters, the SS-MCMG framework precomputes and caches the positional encoding matrix PE at model initialization. Given the maximum sequence length max_len, embedding dimension C_1, position index pos $\in \{0, 1, \ldots, \text{max_len}-1\}$, and channel index $i \in \{0, 1, \ldots, C_1/2-1\}$, the positional encoding on the $2i$ and $2i+1$ channels is calculated as follows:

$$\text{PE}\left(\text{pos}, 2i\right) = \sin\left(\frac{\text{pos}}{10000^{\frac{2i}{C_1}}}\right) \quad (3)$$

$$\text{PE}\left(\text{pos}, 2i+1\right) = \cos\left(\frac{\text{pos}}{10000^{\frac{2i}{C_1}}}\right) \quad (4)$$

where 10 000 is an empirical constant regulating periodicity at different frequencies.

During forward computation, the multiscale convolution feature sequence $S_m \in \mathbb{R}^{B\times L\times C_1}$ is elementwise added with the positional encoding prefix $\text{PE}_{[0:L,:]}$

$$S_m + \text{PE}_{[0:L,:]}. \quad (5)$$

Finally, the sequence with positional information is fed into the multihead self-attention encoder

$$F_m = \text{Transformer}_m\left(S_m + \text{PE}_{[0:L,:]}\right). \quad (6)$$

1) Spectral Probing Module: The SPM, a core component dedicated to understanding the intrinsic spectral properties of substances, is primarily responsible for accurately identifying fine-grained spectral differences of materials in the frequency domain. Each pixel in hyperspectral images contains rich spectral reflection characteristics, and this module serves as the foundation for the model to "precisely recognize spectra and focus on materials."

For input hyperspectral data X, flatten the H and W dimensions

$$X \in \mathbb{R}^{B\times C\times H\times W} \rightarrow x \in \mathbb{R}^{B\times C\times (H\times W)}. \quad (7)$$

Then apply multiscale 1-D parallel convolutions with kernel sizes 1×3, 1×7, and 1×11 to extract spectral features of each pixel in parallel from local to global spectral dimensions. After fusing and mapping the outputs of convolutions at various scales, reshape them into a sequence

$$S_{1\text{-D}} = \text{reshape}\left(\text{Conv}_{1\text{-D}}^{\text{MS}}\left(X\right), \left(B, L, C_1\right)\right), \quad L = H \times W. \quad (8)$$

After elementwise addition with the sine–cosine position encoding $\text{PE}_{[0:L,:]}$, model global dependencies via a multihead self-attention encoder

$$F_{\text{SPM}} = \text{Transformer}_{1\text{-D}}\left(S_{1\text{-D}} + \text{PE}_{[0:L,:]}\right) \quad (9)$$

The schematic of SPM is shown in Fig. 2.

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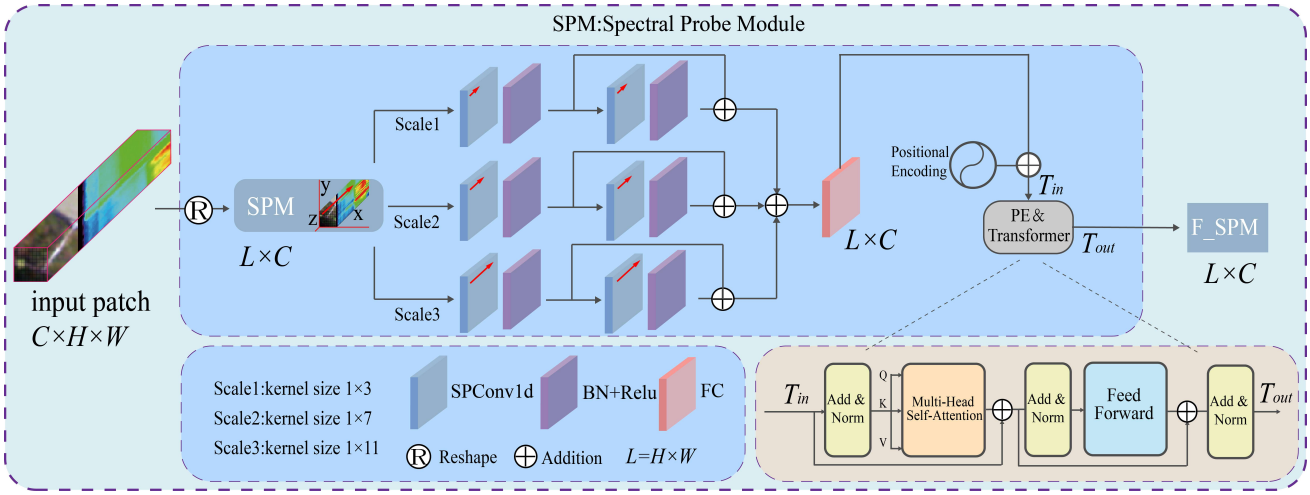


Fig. 2. Flow of SPM: an input patch ($C \times H \times W$) is first reshaped; SPM uses three kernels ($1 \times 3/1 \times 7/1 \times 11$) for multiscale convolution, undergoes BN+ReLU, fully connected layer, and addition to get $L \times C$ features, then via positional encoding and Transformer processing to output F_SPM.

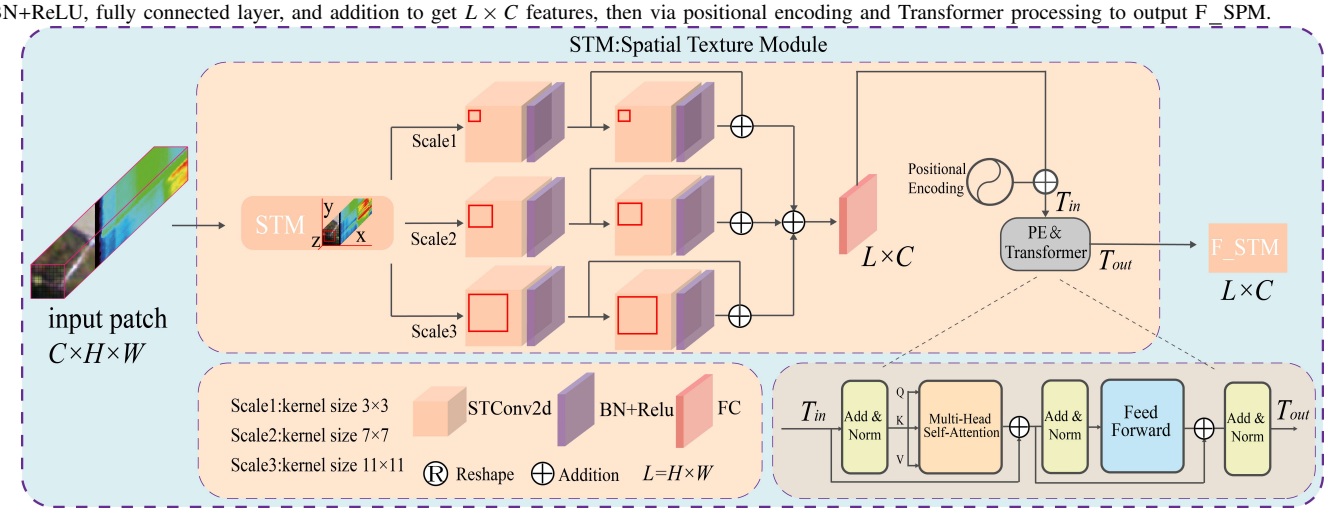


Fig. 3. Flow of the STM: an input patch ($C \times H \times W$) directly enters STM; uses three kernels ($3 \times 3/7 \times 7/11 \times 11$) for multiscale convolution, undergoes BN+ReLU, fully connected layer, and addition to get $L \times C$ features, then via positional encoding and Transformer processing to output F_STM.

2) Spatial Texture Module: The STM focuses on mining the spatial structure and texture information of images, serving as the key component for the model to "perceive shapes and distinguish edges." Designed to capture geometric structures, boundary contours, and spatial distribution features in hyperspectral images, this module applies multiscale 2-D parallel convolutions directly on the original layout of $X \in \mathbb{R}^{B\times C\times H\times W}$, using convolution kernels of 3×3, 7×7, and 11×11 to extract spatial-dimensional features from local-to-global scales. After fusing and mapping the outputs of convolutions at each scale, the result is reshaped into a sequence form

$$S_{2\text{-D}} = \text{reshape}\left(\text{Conv}^{\text{MS}}_{2\text{-D}}(X), (B, L, C_1)\right), \quad L = H \times W. \tag{10}$$

After elementwise addition with the sine–cosine position encoding $\text{PE}_{[0:L,:]}$, global dependencies are modeled through a multihead self-attention encoder

$$F_{\text{STM}} = \text{Transformer}_{2\text{-D}}\left(S_{2\text{-D}} + \text{PE}_{[0:L,:]}\right). \tag{11}$$

The schematic of STM is shown in Fig. 3.

3) Spectral–Spatial Interaction Module (SSIM): The SSIM is the most integrative part among the three modules. It no longer processes frequency domain or spatial information in isolation but organically fuses spectral–spatial relationships through 3-D convolution, serving as a cross-dimensional connector that "understands both content and layout, interprets materials from spectra, and captures space from structures" to achieve integrated modeling. The input hyperspectral data X is first expanded to $X \in \mathbb{R}^{B\times1\times C\times H\times W}$ to ensure that $C \times H \times W$ all undergo convolution operations. The 3-D convolutions with kernel sizes of $3 \times 3 \times 3$, $7 \times 7 \times 7$, and $11 \times 11 \times 11$ are used to extract spectral–spatial information from local-to-global scales.

After fusing and mapping the outputs of convolutions at each scale, the result is restored to $X \in \mathbb{R}^{B\times C\times H\times W}$ and then reshaped into a sequence form

$$S_{3\text{-D}} = \text{reshape}\left(\text{Conv}^{\text{MS}}_{3\text{-D}}(X), (B, L, C_1)\right), \quad L = H \times W. \tag{12}$$

After elementwise addition with the sine–cosine position encoding $\text{PE}_{[0:L,:]}$, global dependencies are modeled through a multihead self-attention encoder

$$F_{\text{SSIM}} = \text{Transformer}_{3\text{-D}}\left(S_{3\text{-D}} + \text{PE}_{[0:L,:]}\right). \tag{13}$$

The schematic of SSIM is shown in Fig. 4.

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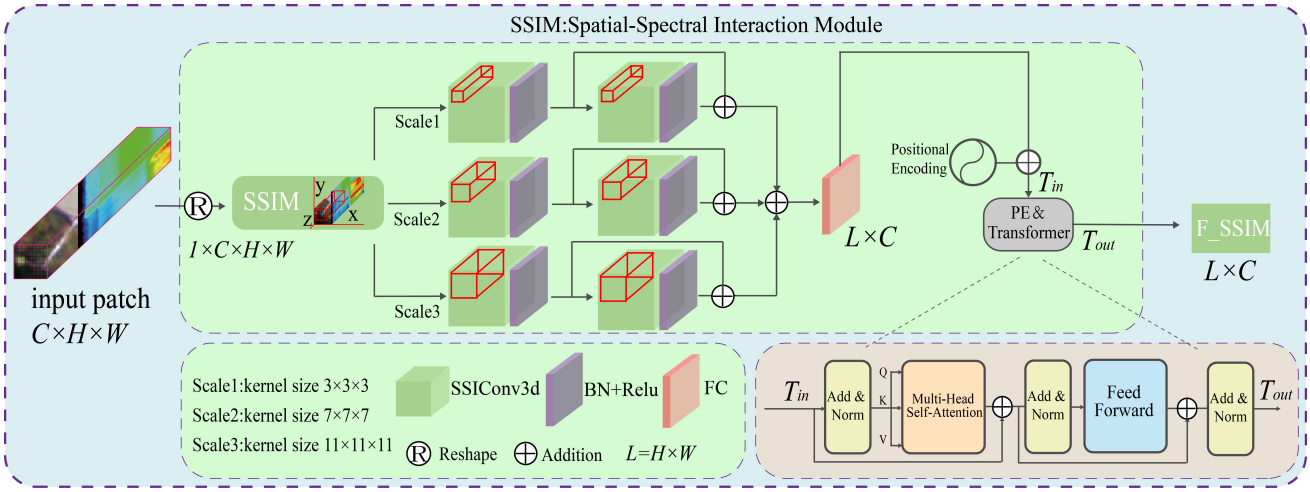


Fig. 4. Flow of the SSIM: an input patch ($C \times H \times W$) is first reshaped to $1 \times C \times H \times W$; SSIM uses three kernels ($3 \times 3 \times 3/7 \times 7 \times 7/11 \times 11 \times 11$) for multiscale convolution, undergoes BN+ReLU, fully connected layer, and addition to get $L \times C$ features, then via positional encoding and Transformer processing to output F_SSIM.

4) Summary: The model outputs three feature substreams (F_{SPM}, F_{STM}, $F_{\text{SSIM}} \in \mathbb{R}^{B\times L\times C_1}$) per batch through three key modules, constructing a universal perceptual modeling paradigm that integrates local fine-grained analysis and global contextual modeling capabilities. The selected convolution kernel sizes in each module ($1 \times 3/1\times7/1\times11$ for SPM, $3 \times 3/7 \times 7/11 \times 11$ for STM, and $3 \times 3 \times 3/7 \times 7 \times 7/11 \times 11\times 11$ for SSIM) are specifically chosen to match the 11×11 patch size adopted in the model design. This ensures full coverage of local regions while enabling hierarchical feature extraction from fine-to-coarse granularity. The multi-scale configuration enhances the model's adaptability to spatial structures of various sizes and spectral variation patterns across frequency bands.

The next step will involve unified integration of these three types of features via SS-MCAF to enhance the overall representation ability.

B. Spatial–Spectral Multidimensional Collaborative Attention Fusion

Given the features F_{SPM}, F_{STM}, and F_{SSIM} received from SS-MCMG, to construct the collaborative relationships among queries, keys, and values, we first concatenate pairs of features along the feature dimension to form key–value pairs for the collaborative attention computation

$$\text{KV}_{\text{SPM}} = [F_{\text{STM}}, F_{\text{SSIM}}] \in \mathbb{R}^{B\times L\times 2C_1} \tag{14}$$

$$\text{KV}_{\text{STM}} = [F_{\text{SPM}}, F_{\text{SSIM}}] \in \mathbb{R}^{B\times L\times 2C_1} \tag{15}$$

$$\text{KV}_{\text{SSIM}} = [F_{\text{SPM}}, F_{\text{STM}}] \in \mathbb{R}^{B\times L\times 2C_1} \tag{16}$$

where $[\cdot,\cdot]$ denotes the concatenation along the feature dimension, expanding the dimension from (B, L, C_1) to $(B, L, 2C_1)$. To maintain consistency in the feature dimension, a linear projection layer $W_{\text{proj}} \in \mathbb{R}^{2C_1\times C_1}$ is applied to map the concatenated features back to the original feature dimension C_1

$$\text{KV}_i = W_{\text{proj}}\text{KV}_i, \quad i \in \{\text{SPM}, \text{STM}, \text{SSIM}\}. \tag{17}$$

Next, employing the multihead self-attention mechanism, each of F_{SPM}, F_{STM}, and F_{SSIM} is used as a query to compute the attention interaction with their corresponding key–value pairs

$$\text{MF}_{\text{SPM}} = \text{MultiHead}\,(F_{\text{SPM}}, \text{KV}_{\text{SPM}}, \text{KV}_{\text{SPM}}) \tag{18}$$

$$\text{MF}_{\text{STM}} = \text{MultiHead}\,(F_{\text{STM}}, \text{KV}_{\text{STM}}, \text{KV}_{\text{STM}}) \tag{19}$$

$$\text{MF}_{\text{SSIM}} = \text{MultiHead}\,(F_{\text{SSIM}}, \text{KV}_{\text{SSIM}}, \text{KV}_{\text{SSIM}}) \tag{20}$$

where $\text{MultiHead}(Q, K, V)$ denotes the standard multihead self-attention operation. The resulting three fused feature sets MF_{SPM}, MF_{STM}, and MF_{SSIM} contain collaboratively integrated information from different modules.

Finally, to further enhance the fusion effect, the three feature sets are concatenated along the feature dimension and then mapped through a fully connected layer to obtain the final fused feature

$$\text{MF}_{\text{fused}} = W_{\text{fc}}\,[\text{MF}_{\text{SPM}}, \text{MF}_{\text{STM}}, \text{MF}_{\text{SSIM}}] \tag{21}$$

where W_{fc} is a linear transformation layer that projects the fused feature back to the original feature dimension C. The output MF_{fused} serves as the output for this module.

C. Consistency Loss and Diversity Loss

After multidimensional collaborative attention fusion, to further optimize the feature representation capability, this work introduces a CL and a DL to maintain information stability among features from different modules and enhance feature discriminability.

1) Consistency Loss: The goal of the CL is to strengthen the similarity among features from different modules, ensuring that features belonging to the same category remain consistent across modules. Specifically, the mean squared error (mse) between features of different modules is calculated as follows:

$$\begin{aligned} L_{\text{consistency}} = \frac{1}{3}\big(&\|\text{MF}_{\text{SPM}} - \text{MF}_{\text{STM}}\|^2 \\ &+ \|\text{MF}_{\text{SPM}} - \text{MF}_{\text{SSIM}}\|^2 \\ &+ \|\text{MF}_{\text{STM}} - \text{MF}_{\text{SSIM}}\|^2\big) \end{aligned} \tag{22}$$

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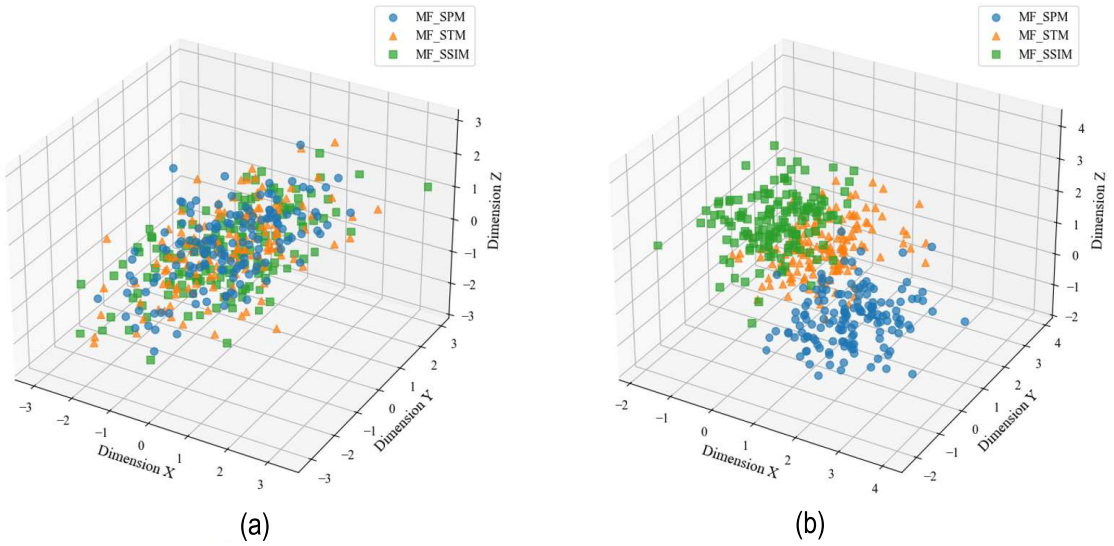


Fig. 5. 3-D spatial distribution comparison of module features. Both illustrating features from the SPM (blue dots), STM (orange triangles), and spatial–spectral interaction module (green squares). (a) 3-D distribution of features before loss function optimization. (b) Corresponding distribution after optimization.

where $\mathrm{MF}_{\mathrm{SPM}}$, $\mathrm{MF}_{\mathrm{STM}}$, and $\mathrm{MF}_{\mathrm{SSIM}}$ represent the fused features obtained from the SPM, STM, and spatial–spectral interaction module, respectively. This loss constrains the Euclidean distances among features to ensure their distributions in the feature space are close, thereby improving model stability.

2) Diversity Loss: To avoid excessive feature convergence among different modules leading to redundant information, a DL is introduced to maximize the dissimilarity among features from different modules. Cosine similarity is adopted to measure the similarity between features, and this similarity is minimized

$$\cos_{\mathrm{sim}}(A,B) = \frac{A \cdot B}{\|A\|\|B\| + \epsilon} \tag{23}$$

$$\begin{aligned} L_{\mathrm{diversity}} = \frac{1}{3}\big(&\cos_{\mathrm{sim}}(\mathrm{MF}_{\mathrm{SPM}}, \mathrm{MF}_{\mathrm{STM}}) \\ &+ \cos_{\mathrm{sim}}(\mathrm{MF}_{\mathrm{SPM}}, \mathrm{MF}_{\mathrm{SSIM}}) \\ &+ \cos_{\mathrm{sim}}(\mathrm{MF}_{\mathrm{STM}}, \mathrm{MF}_{\mathrm{SSIM}})\big) \end{aligned} \tag{24}$$

where ϵ is a small constant added to prevent numerical instability. The DL encourages independence among features from different modules by minimizing their cosine similarity, thus enabling the learning of complementary information.

3) Adaptive Loss Weighting: Since there may be a tradeoff between the CL and the DL during optimization, a dynamic weighting mechanism is further introduced to adaptively balance their relative contributions

$$\alpha = \exp\left(L_{\mathrm{consistency}}\right), \quad \beta = \exp\left(-L_{\mathrm{diversity}}\right) \tag{25}$$

$$\bar{\alpha} = \frac{\alpha}{\alpha+\beta}, \quad \bar{\beta} = \frac{\beta}{\alpha+\beta} \tag{26}$$

$$L_{\mathrm{final}} = \bar{\alpha} L_{\mathrm{consistency}} + \bar{\beta} L_{\mathrm{diversity}}. \tag{27}$$

Here, the exponential form of α ensures that the weight of the CL increases with its magnitude, while the negative exponential form of β guarantees that the weight of the DL increases as its value decreases. This design allows the model to focus on whichever objective is currently more under-optimized. In other words, when module features are inconsistent, the CL becomes larger, leading to a larger α and thus a greater focus on alignment. Conversely, when feature diversity is insufficient, the DL is small, resulting in a higher β, which pushes the model to enhance distinctiveness.

This adaptive mechanism enables the network to dynamically balance the two objectives throughout training, preventing either one from dominating and thereby improving the robustness of feature representation learning.

To intuitively demonstrate this effect, Figs. 5 and 6 are provided. Fig. 5 shows the comparison of the 3-D projected distributions of module features, color-coded by class. The left plot shows distributions without loss constraints—where class boundaries are unclear—while the right plot (with loss constraints) exhibits tighter intraclass clustering and clearer interclass separation, highlighting the benefit of the proposed losses. Fig. 6 illustrates the adaptive weight trends across 100 training epochs: the CL (blue) decreases and stabilizes over time, indicating growing feature alignment, while the DL (orange) increases and fluctuates, reflecting the emergence of diverse representations. The corresponding weights (green and red curves) shift accordingly, confirming the model's ability to adjust focus at different optimization stages.

III. Dataset Construction

A. UAV-Based Hyperspectral Remote Sensing Data Acquisition System

The UAV hyperspectral remote sensing data acquisition system used in this study consists of a UAV platform and an onboard hyperspectral imaging sensor, as shown in Fig. 7. The UAV platform adopts a multirotor design, capable of both hovering and executing planned flight paths. The hyperspectral imaging sensor is a dedicated airborne device mounted on a two-axis gimbal, enabling image stabilization and vibration reduction during data acquisition.

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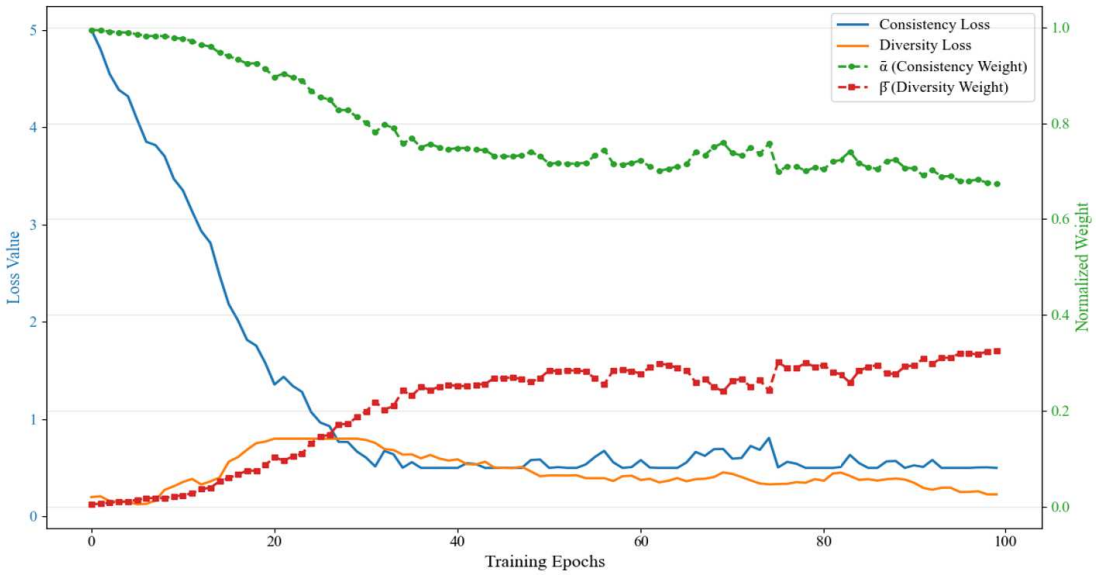


Fig. 6. Adaptive weight variation trend shows the changes in key metrics over 100 training epochs. The x-axis represents training epochs, the left y-axis denotes loss values, and the right y-axis indicates normalized weights. The blue curve represents the CL, the orange curve represents the DL, the green curve represents the consistency weight (δ), and the red curve represents the diversity weight (β).



Fig. 7. UAV-based hyperspectral remote sensing data acquisition system. a—UAV platform, b—control handle, c—hyperspectral gimbal.

TABLE I
MAIN SPECIFICATIONS OF THE UAV PLATFORM

Categories	Parameter
Dimensions	810 × 670 × 430 mm (L × W × H)
Wheelbase	895 mm
Max Takeoff Weight	9 kg
Max Payload	2.7 kg
RTK Position Accuracy	1 cm + 1 ppm (horizontal); 1.5 cm + 1 ppm (vertical)
Max Horizontal Speed	23 m/s (S-mode); 17 m/s (P-mode)
Max Flight Altitude	5000 m–7000 m
Max Wind Resistance	12 m/s
Max Flight Time	55 min

1) UAV Platform: The UAV platform employed in this experiment is the Matrice 300 RTK multirotor drone developed by DJI Innovations, Shenzhen, China, as shown in Fig. 3(a). The UAV system comprises four main components: power system, flight control system, avionics system, and airframe structure. The main technical specifications of UAV are listed in Table I.

TABLE II
MAIN SPECIFICATIONS OF THE X20P AIRBORNE HSI IMAGER

Categories	Parameter
Dimensions	169 mm × 145 mm × 172 mm
Structural Weight	Integrated gimbal design, total weight $<$ 1.5 kg
Spectral Range	350–1000 nm
Spectral Resolution	4 nm
Number of Bands	164
Hyperspectral Resolution	1886 × 1886 pixels/frame
Hyperspectral Imaging Speed	$>$ 2 Cubes/s (1886 × 1886 pixels per cube)
Detector Size	12.16 × 12.16 mm
Focal Length	19 mm
Optical Array / Field of View	66 elements / 35°

2) Airborne HSI Imager: The onboard hyperspectral sensor used in this study is the X20P hyperspectral imaging device manufactured by Cubert GmbH. The X20P is based on light-field imaging technology and features a 20-MP ultrahigh-definition CMOS sensor. It adopts a snapshot imaging approach to rapidly capture hyperspectral images with more than 160 spectral bands, continuously covering the wavelength range from 350 to 1000 nm. With dual-GigE camera interfaces, it supports an image frame rate of up to 5 Hz (1886 × 1886 pixels/frame). The technical specifications of the hyperspectral sensor are shown in Table II.

B. Study Area

In this study, the Yellow River Wetland Park located in Huiji, Zhengzhou, Henan, China, was selected as the research area, as shown in Fig. 8. This region lies within

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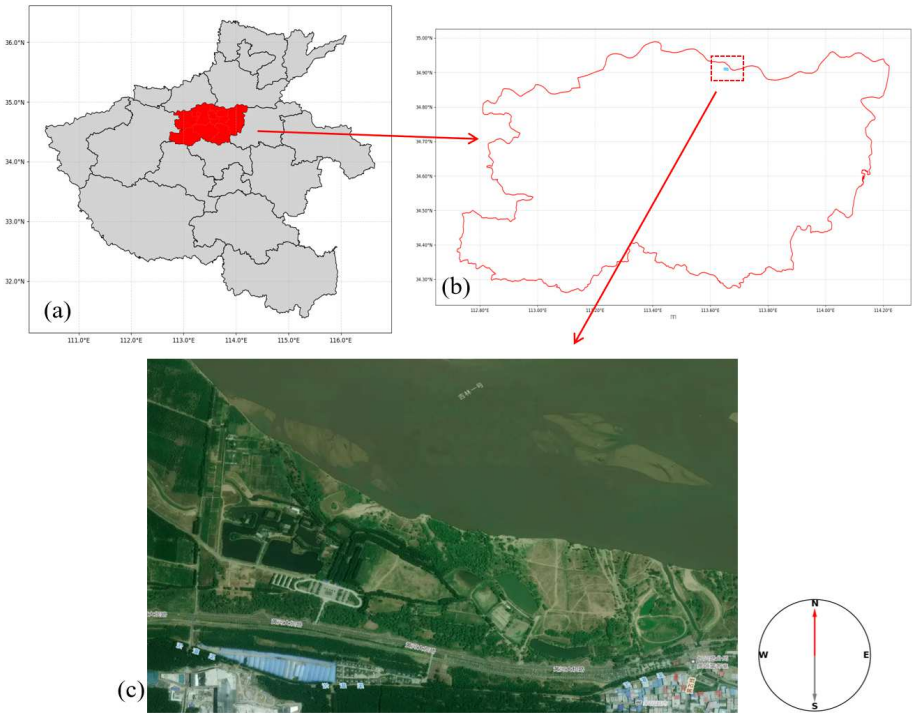


Fig. 8. Research area. (a) Administrative map of Henan with Zhengzhou highlighted in red. (b) Boundary of Zhengzhou City with the wetland park area is marked by a dashed rectangle. (c) Satellite image of the park with a compass indicating orientation in the lower right corner.

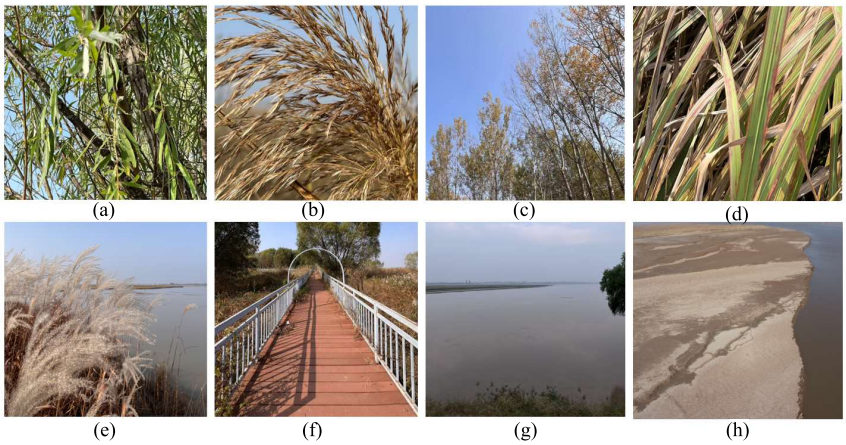


Fig. 9. Field-surveyed examples of land-cover categories. (a) Willow, (b) Reed, (c) Poplar, (d) Cogon Grass, (e) Miscanthus, (f) Road, (g) Yellow River, and (h) Mudflat.

the Yellow River Basin, near the coordinates of longitude 113°39′37.38″ E and latitude 34°55′3.97″ N, and represents a typical wetland ecosystem. It was officially designated as a National Wetland Park in 2015 [45]. To ensure the scientific validity and representativeness of the data, a typical sample plot featuring complex land-cover types and diverse vegetation was selected for investigation. Field surveys and vegetation classification were conducted accordingly. The main land-cover types are illustrated in Fig. 9.

C. Presentation of Our Constructed Dataset

The dataset used in this study includes both the original hyperspectral data and the high-resolution data obtained through fusion with a panchromatic image. The hyperspectral camera captured rich spectral information, though its spatial resolution is limited by hardware constraints. In contrast, the panchromatic camera provides high-resolution imagery in a single spectral band. To enhance spatial resolution while preserving spectral characteristics, this study employs a pan-sharpening technique to fuse the hyperspectral data with the panchromatic image. A false-color representation of the data is shown in Fig. 10. To construct the hyperspectral image classification dataset, eight representative land-cover categories were identified based on ground survey records as classification targets. Annotation was performed on the hyperspectral images, where experts manually selected regions of interest (ROIs) and assigned class labels according to the results of on-site investigations, ensuring the accuracy and reliability of the labeled data. For dataset partitioning, a random sampling strategy was adopted, in which 5% of the labeled samples were allocated to the training set and the remaining 95% to

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(a) (b)

Fig. 10. False-color data: both images were captured by a UAV at a flight altitude of 400 m. (a) False-color raw hyperspectral data with dimensions of (330, 330, 150) (height × width × bands), with a data size of approximately 62.5 MB. The displayed bands correspond to R (638 nm), G (550 nm), and B (470 nm), and the data are stored in floating-point format. (b) False-color fused hyperspectral data with dimensions of (1500, 1500, 150), with a data size of approximately 1.26 GB. The displayed bands are the same as those in the original data [R (638 nm), G (550 nm), B (470 nm)] and are also stored in floating-point format.

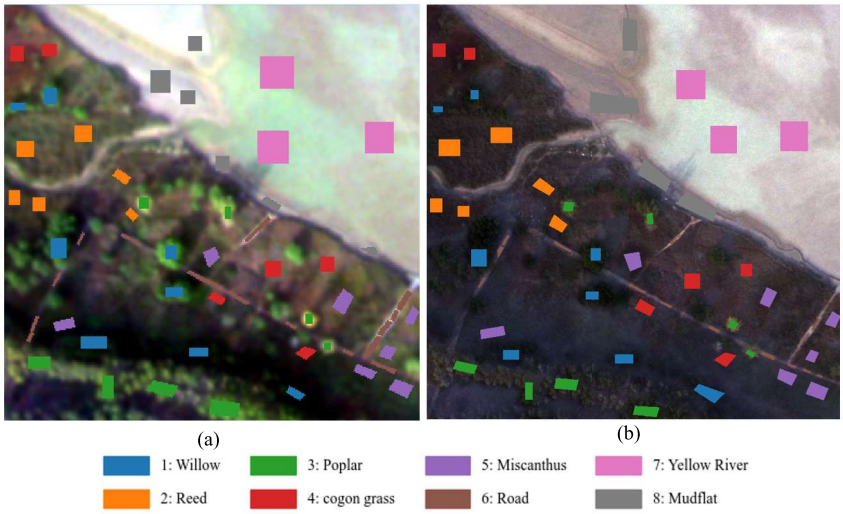


Fig. 11. ROI annotations. (a) ROI annotation map of the pseudo-color original hyperspectral data. (b) ROI annotation map of the pseudo-color fused hyperspectral data. The bottom legend indicates the corresponding land-cover types represented by specific label colors.

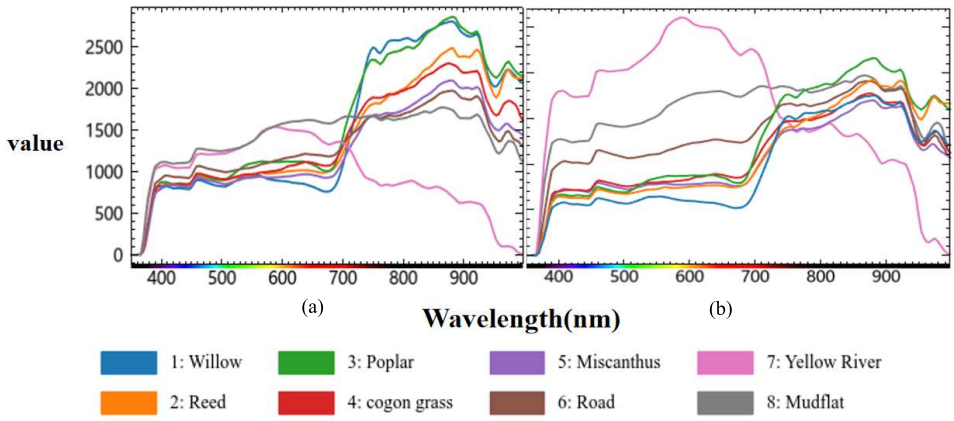


Fig. 12. Visualization of mean spectral curves for each class. (a) Spectral curves of the original hyperspectral data. (b) Spectral curves of the fused data. The *x*-axis represents wavelength (nanometers), and the *y*-axis represents spectral reflectance values. The bottom legend indicates the corresponding land-cover types represented by specific label colors.

the testing set. This design reflects real-world scenarios where limited training data is available, allowing for a more practical evaluation of the model's generalization capability. During the selection of training samples, the class balance was maintained

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TABLE III
SAMPLE NUMBERS OF TRAINING AND TESTING SETS. (A) ORIGINAL HYPERSPECTRAL DATA. (B) FUSED HYPERSPECTRAL DATA

(a)			
Class No	Class Name	Train	Test
1	Willow	46	880
2	Reed	40	766
3	Poplar	42	796
4	Cogon Grass	34	663
5	Miscanthus	44	838
6	Road	16	300
7	Yellow River	57	1087
8	Mudflat	22	436
Total		**291**	**5991**
(b)			
Class No	Class Name	Train	Test
1	Willow	990	18810
2	Reed	787	14963
3	Poplar	923	17528
4	Cogon Grass	742	14108
5	Miscanthus	934	17741
6	Road	146	2779
7	Yellow River	1215	23085
8	Mudflat	450	8550
Total		**6187**	**117513**

TABLE IV
PROPORTION OF EACH CLASS IN THE WHOLE IMAGE

Class No	Class Name	Fusion Data	Original Data
1	Willow	0.88%	0.85%
2	Reed	0.70%	0.74%
3	Poplar	0.82%	0.77%
4	Cogon Grass	0.66%	0.64%
5	Miscanthus	0.83%	0.81%
6	Road	0.13%	0.12%
7	Yellow River	1.08%	1.05%
8	Mudflat	0.40%	0.42%
Total Labeled		**5.4**	**5.3**

to minimize performance bias caused by data imbalance. The ROI annotations are illustrated in Fig. 11. The number of training and testing samples is listed in Table III, where (a) presents the sample counts for the original hyperspectral data, and (b) shows those for the fused data. To ensure that each class corresponds to its true spatial location, the annotation process for the original and fused images was conducted independently but strictly based on field investigation data. Although there are slight visual differences in the labeled areas due to separate annotation, each region precisely corresponds to its designated class without any misalignment or class confusion. To further illustrate the consistency of annotation, we calculated the spatial proportion of each class in both images and included the results in this article, as shown in Table IV. The results show that the spatial proportions of the classes in both images differ only slightly, indicating well-controlled annotation quality and reinforcing the overall accuracy (OA) and reliability of the dataset.

Average Spectral Curves: By calculating the mean spectra of each land-cover class and visualizing them, the average spectral curves are plotted in Fig. 12. Fig. 12(a) shows the spectral curves from the original hyperspectral data. Fig. 12(b) presents those from the fused data. It can be observed that Cogon Grass, Miscanthus, and Reed exhibit similar spectral characteristics, which may cause confusion during classification. In addition, in the original data, the spectral curve of road shares similarity with the aforementioned three vegetation types across multiple bands. Furthermore, willow and poplar show similar spectral variation trends beyond 700 nm, whereas mudflat and the Yellow River have comparable spectral responses below 600 nm. Compared with the original data, the fused hyperspectral data demonstrates stronger discriminative capability among classes. Particularly, the spectral features of the road, Yellow River, and mudflat are clearer with more distinct boundaries. Although willow and poplar still have similar spectral shapes, their reflectance values differ significantly across multiple bands. Overall, the fused data hold greater potential for improving classification accuracy. Meanwhile, the interclass confusion present in the original hyperspectral data offers a higher challenge for model training and feature extraction, which aids in thoroughly evaluating the robustness and discriminative power of algorithms.

IV. EXPERIMENTS

A. Experimental Setup

All experiments were conducted using a GeForce RTX 4090 GPU and implemented with PyTorch version 2.6.0+cu126. To ensure the reliability and consistency of the results, the number of training epochs for the original hyperspectral data was set to 200. The batch size and block size were fixed at 256 and 11, respectively. Model training was performed using the Adam optimizer with an initial learning rate of 0.001 and a weight decay of $1e^{-4}$ to prevent overfitting. In addition, the core Transformer module was configured as follows: the Transformer encoder consists of two layers, each with 16 attention heads, and the embedding dimension is set to 150. To reduce the impact of randomness, each model was independently trained five times, and the average performance across these runs was reported as the final result. To address the issue of limited training samples, data augmentation techniques were applied during training, including random rotations, horizontal and vertical flips, and the addition of Gaussian noise, to enhance the model's robustness and generalization ability.

B. Evaluation Metrics

In addition to evaluating classification accuracy for each individual class, three quantitative metrics were used to analyze and assess model performance: OA, average accuracy (AA), and kappa coefficient. Higher values of per-class accuracy, OA, AA, and kappa indicate better classification performance of the model.

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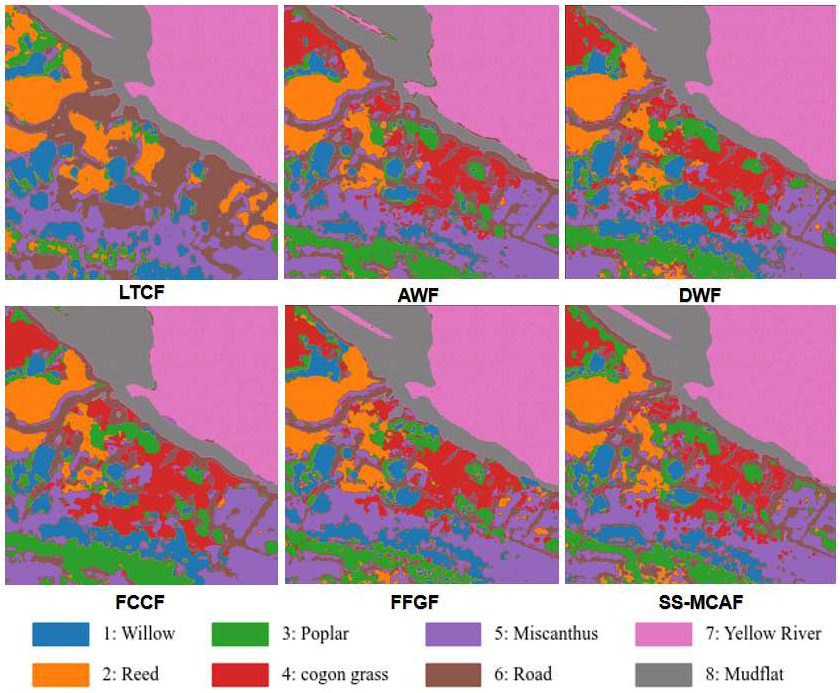


Fig. 13. Classification results of comparative experiments on feature fusion. This figure visualizes the classification outcomes of different feature fusion models (LTCF, AWF, DWF, FCCF, FFGF, and SS-MCAF) over the study area. The bottom legend indicates the corresponding land-cover types represented by specific label colors.

TABLE V
ABLATION EXPERIMENTAL TABLE

STM	SPM	SS-MCAF	CL	DL	OA
×	×	×	×	×	83.14%
√	√	×	×	×	89.28%
√	√	√	×	×	95.57%
√	√	√	×	√	96.18%
√	√	√	√	×	97.12%
√	√	√	√	√	98.08%

C. Ablation Study

This study designs a three-branch separated modeling structure, specifically extracting STM, SPM, and SSIM. Although the SSIM utilizes multiscale 3-D convolutions to jointly model spectral and spatial information and possesses certain expressive capability, its structure struggles to accurately distinguish fine-grained spatial textures from high-frequency spectral distribution features, which may cause information confusion and limit its ability to capture heterogeneous multidimensional features. In contrast, STM and SPM adopt dimension-specific designs, enabling the model to more effectively and efficiently focus on critical features within their respective domains, thus achieving more specialized and complementary feature representations. The accuracy values mentioned below are all OA results.

To verify whether a single, more powerful SSIM module can replace the three-branch separated design, we first trained the model using only the SSIM, resulting in an OA of 83.14% on the testing set. Under conditions without complex fusion mechanisms, simply introducing STM and SPM with basic weighted fusion significantly improved accuracy to 89.28%, clearly demonstrating that multidimensional features have significant differences and complementarity in information types, and that unified modeling via a single path tends to cause feature conflicts and degrade performance. Further employing the specially designed SS-MCAF module to collaboratively fuse the three feature streams raised accuracy to 95.57%. By additionally integrating CL and DL to enhance intermodule collaboration and complementarity, the accuracy was ultimately improved to 98.08%. These results strongly demonstrate that explicitly separating the three feature streams and employing a collaborative fusion strategy is a rational design based on the inherent nature of multimodal heterogeneous features and is far superior to unified modeling with a single powerful module. The ablation experiment results are summarized in Table V.

D. Comparative Experiments on Feature Fusion

To evaluate the effectiveness of feature fusion strategies in classification tasks, a series of comparative experiments was conducted based on the original hyperspectral data. All methods employed SS-MCMG for feature modeling. The compared strategies include: LTCF [39], AWF [40], DWF [41], FCCF [42], and FFGF [43].

Experimental results demonstrate that the proposed SS-MCAF outperforms all baseline fusion strategies across multiple evaluation metrics, confirming the superiority and versatility of the proposed fusion mechanism. The classification visualization results are shown in Fig. 13, and the detailed classification accuracies are presented in Table VI. Although the original hyperspectral images suffer from low spatial resolution and suboptimal image quality, this exper-

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TABLE VI
CLASSIFICATION ACCURACY COMPARISON OF DIFFERENT FEATURE FUSION METHODS

Class / Method	LTCF [39]	AWF [40]	DWF [41]	FCCF [42]	FFGF [43]	SS-MCAF
1.Willow	91.35%	93.00%	98.04%	92.79%	91.66%	**96.40%**
2.Reed	96.85%	99.86%	99.75%	99.45%	98.89%	**99.86%**
3.Poplar	30.05%	96.99%	96.89%	92.85%	89.64%	**99.79%**
4.Cogon grass	0.00%	84.87%	91.35%	88.62%	87.75%	**96.40%**
5.Miscanthus	39.18%	91.75%	97.02%	90.95%	84.54%	**98.63%**
6.Road	87.63%	91.11%	85.89%	93.03%	87.63%	**88.33%**
7.Yellow River	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
8.Mudflat	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
OA	71.92%	95.59%	97.08%	95.44%	93.62%	**98.08%**
AA	68.13%	94.70%	96.15%	94.78%	92.65%	**97.42%**
Kappa	67.29%	94.85%	96.58%	94.68%	92.54%	**97.75%**

TABLE VII
CLASSIFICATION ACCURACY COMPARISON OF DIFFERENT MODELS

Class / Method	3D-CNN [32]	SSRN [34]	CMR-CNN [35]	DPRnet [36]	MCVT [37]	SS-MCNet
1.Willow	95.25%	94.57%	96.81%	92.69%	93.31%	**96.40%**
2.Reed	97.87%	96.81%	98.98%	96.28%	93.88%	**99.86%**
3.Poplar	89.51%	88.21%	90.33%	93.16%	98.03%	**99.79%**
4.Cogon grass	75.65%	72.33%	79.39%	44.67%	77.46%	**96.40%**
5.Miscanthus	85.55%	83.54%	86.80%	87.06%	79.73%	**98.63%**
6.Road	86.04%	84.28%	88.07%	96.21%	92.56%	**88.33%**
7.Yellow River	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
8.Mudflat	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
OA	92.35%	91.54%	93.73%	91.18%	93.11%	**98.08%**
AA	91.21%	90.32%	92.55%	88.76%	91.87%	**97.42%**
Kappa	90.18%	89.34%	91.49%	87.15%	90.71%	**97.75%**

iment conducted a fair evaluation of all methods under the same training epochs (200 epochs) to improve the model stability. The results indicate that traditional methods exhibit unstable performance across multiple classes, with significant discrepancies. For example, LTCF achieved recognition accuracies of only 0.00% and 39.18% for the "Cogon Grass" and "Miscanthus" classes, respectively, demonstrating its notable limitations in handling spectrally similar categories. AWF showed considerable fluctuations in classes such as "Willow" and "Road," reflecting its uneven class discrimination capability. In contrast, the proposed SS-MCNet achieved stable and efficient recognition across all eight classes, with an OA of 98.08%, AA of 97.42%, and a kappa coefficient of 97.75%. Notably, it demonstrated significant advantages in easily confused classes such as "Willow" and "Poplar," as well as "Cogon Grass" and "Miscanthus," fully reflecting its robustness and strong discriminative ability on low-quality hyperspectral data.

E. Comparative Experiments on Feature Modeling

To evaluate the model classification performance, this experiment was carried out based on the original hyperspectral data, selecting representative hyperspectral classification models in recent years for comparison, including 3-D-CNN [32], SSRN [34], CMR-CNN [35], DPRnet [36], MCVT [37], and SS-MCNet proposed in this article. Based on a unified input size, this study evaluated the complexity of various comparison models, showing that 3-D-CNN has approximately 60k parameters and 432 GFLOPs; SSRN has 130k parameters and 502 GFLOPs; CMR-CNN has 170k parameters and 616 GFLOPs; DPRNet has 120k parameters and 478 GFLOPs; MCVT has 210k parameters and 702 GFLOPs; while the proposed SS-MCNet has about 80k parameters and 519 GFLOPs.

The classification visualization results are shown in Fig. 14, and the detailed classification accuracies are presented in Table VII. From the classification results, SS-MCNet performed excellently in all categories, especially demonstrating significant advantages in easily confused categories. Taking Willow and Poplar as examples, the accuracies of CMR-CNN and SSRN for Poplar were 98.03% and 88.21%, respectively, while SS-MCNet increased this to 99.79%. In spectrally similar vegetation categories such as Reed, Cogon Grass, and Miscanthus, DPRnet only achieved recognition accuracies of 44.67% for Cogon Grass and 87.06% for Miscanthus,

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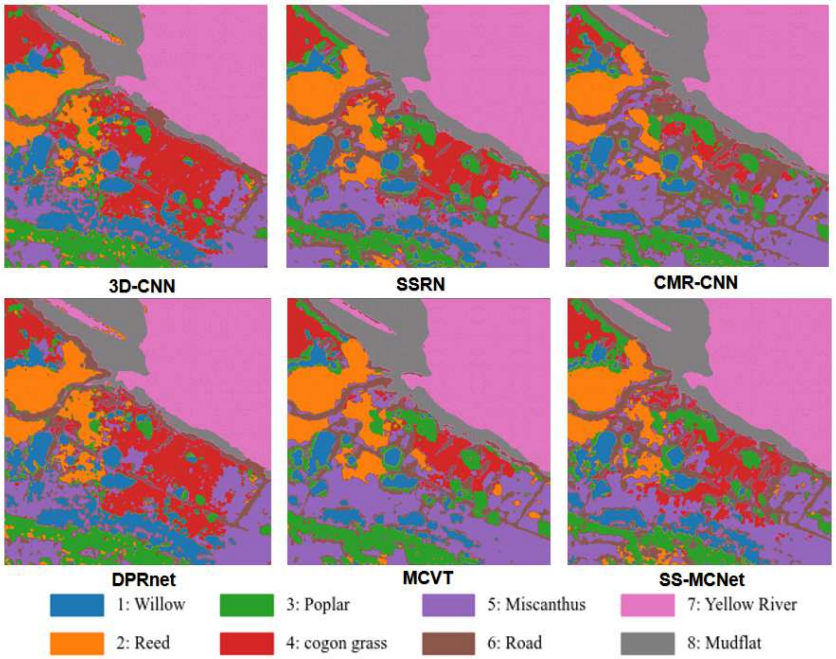


Fig. 14. Classification results comparison of different models. Visualization of classification results for the study area using various models (3-D-CNN, SSRN, CMR-CNN, DPRnet, MCVT, and SS-MCNet). The bottom legend indicates the corresponding land-cover types represented by specific label colors.

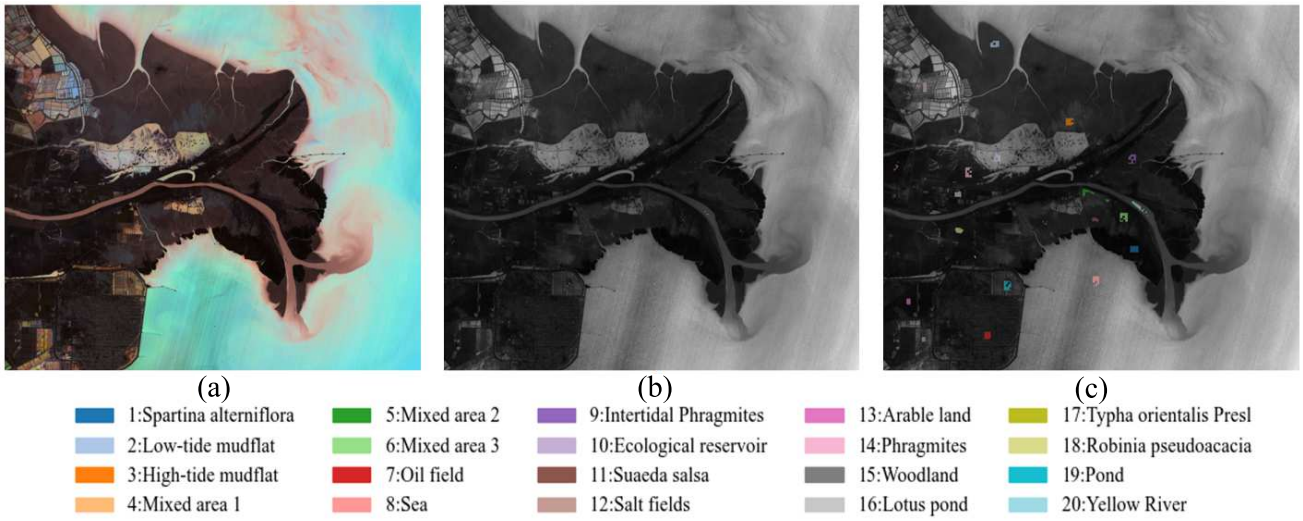


Fig. 15. False-color data and ROI annotations. (a) False-color raw hyperspectral data with dimensions of (1185, 1342, 285) (height × width × bands), with a data size of approximately 2.71 GB; the displayed bands correspond to R (640 nm), G (550 nm), and B (470 nm), and the data are stored in floating-point format. (b) ROI distribution of the hyperspectral data training set. (c) ROI distribution of the hyperspectral data testing set. The bottom legend indicates the corresponding land-cover types represented by specific label colors.

whereas SS-MCNet reached 96.40% and 98.63%, respectively. MCVT also showed unsatisfactory performance in these categories. For categories with distinct spectral features like Yellow River and Mudflat, all models achieved a 100% recognition rate, indicating that model differences mainly lie in the discrimination of complex categories. Comprehensive evaluation indicators show that SS-MCNet achieved 98.08% in OA, 97.42% in AA, and 97.75% in the kappa coefficient, representing improvements of 6.90%, 8.66%, and 10.6 percentage points compared with DPRnet. The experimental results confirm that SS-MCNet outperforms existing comparative methods in both fine-category discrimination and overall classification performance.

F. Experiments on an Open-Source Dataset

To verify that the proposed model performs well not only on the self-constructed dataset but also on public datasets, we further conducted experiments using an open-source dataset. This dataset was acquired by the GF-5 satellite over the Yellow River Delta region. This dataset is named the Yellow River Estuary dataset. The image size is 1185 × 1342 pixels with a spatial resolution of 30 m. It contains 285 spectral bands and includes 20 land-cover types. Table VIII lists the number of training and testing samples for each class as well as the total. Fig. 15 shows the pseudo-color image and sample distribution map. To evaluate the classification performance of the model, experiments were conducted on the original

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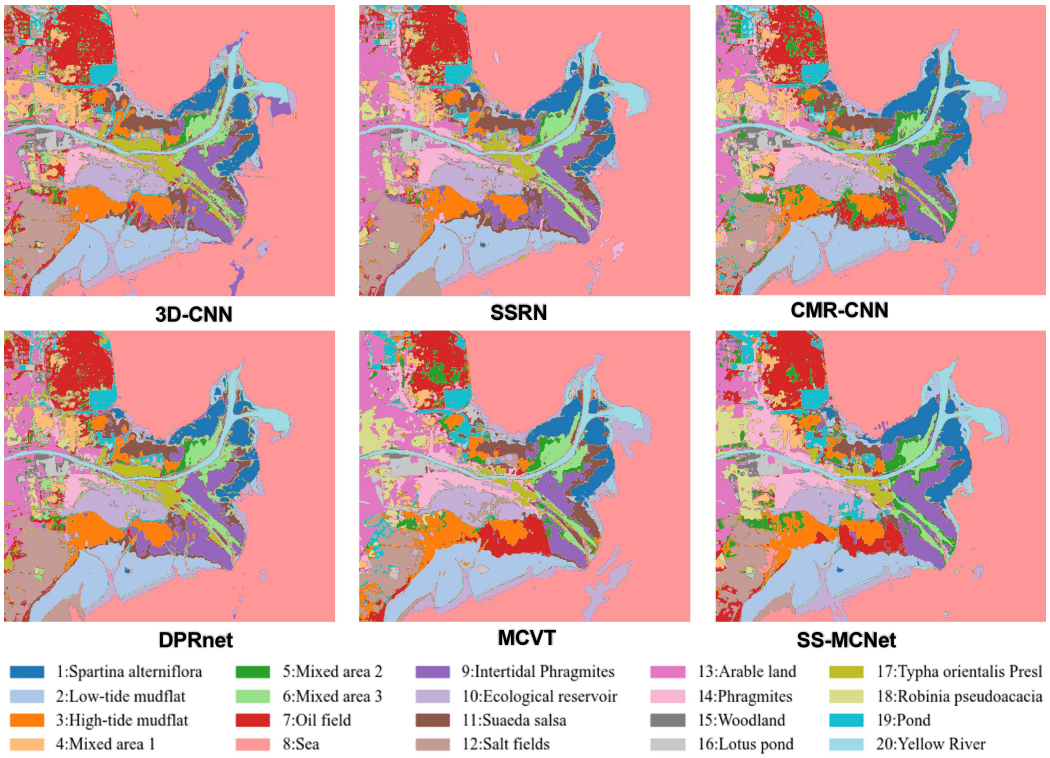


Fig. 16. Comparison of classification results on the Yellow River Estuary datasets. Visualization of classification results for the study area using various models (3-D-CNN, SSRN, CMR-CNN, DPRnet, MCVT, and SS-MCNet). The bottom legend indicates the corresponding land-cover types represented by specific label colors.

TABLE VIII
SAMPLE NUMBERS OF TRAINING AND TESTING SETS

Yellow River Estuary			
Class No	Class Name	Train	Test
1	Spartina alterniflora	84	743
2	Low-tide mudflat	84	709
3	High-tide mudflat	91	645
4	Mixed area 1	16	123
5	Mixed area 2	95	774
6	Mixed area 3	75	731
7	Oil field	72	612
8	Sea	135	756
9	Intertidal Phragmites	86	758
10	Ecological reservoir	75	563
11	Suaeda salsa	67	610
12	Salt fields	81	663
13	Arable land	68	681
14	Phragmites	83	756
15	Woodland	49	502
16	Lotus pond	75	461
17	Typha orientalis Presl	6	30
18	Robinia pseudoacacia	65	516
19	Pond	86	763
20	Yellow River	88	771
Total		**1481**	**12167**

hyperspectral data. We compared the results with several representative hyperspectral classification models proposed in recent years, including 3-D-CNN [32], SSRN [34], CMR-CNN [35], DPRnet [36], MCVT [37], and SS-MCNet proposed in this article. The classification visualization results are shown in Fig. 16, and the accuracy metrics are presented in Table IX.

A brief analysis of the classification accuracy reveals that the proposed SS-MCNet consistently achieves the best performance across the majority of land-cover classes. Notably, it shows clear advantages in difficult categories such as Class 5 (Mixed area 2), Class 11 (Suaeda salsa), and Class 16 (Lotus pond), where other models struggle to achieve satisfactory accuracy. In addition to its strong per-class performance, SS-MCNet also achieves the highest OA: 97.56%, AA: 95.43%, and kappa coefficient: 97.21%. This experiment demonstrates that the proposed model not only achieves superior classification performance on the self-constructed UAV dataset but also generalizes well to the open-source Yellow River Estuary dataset, confirming its robustness and effectiveness across diverse hyperspectral classification tasks.

G. Comparative Experiment Between Fusion Data and Original Data

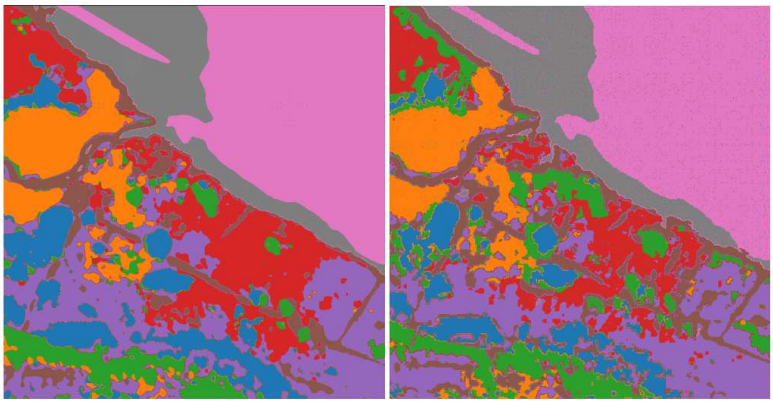
Given that the fused dataset contains richer spatial information and a significantly larger data volume than the original hyperspectral data, practical measurements indicate that training the fusion data for approximately 20 epochs requires a similar amount of time as training the original data for 200 epochs. Therefore, 20 training epochs were used for the fusion data in the comparative experiments.

As shown in Fig. 17 and Table X, the classification results reveal that the fusion and original data each exhibit advantages in different categories. For example, the fusion data achieves

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TABLE IX
CLASSIFICATION ACCURACY COMPARISON OF YELLOW RIVER ESTUARY DATASET

Class / Method	3D-CNN [32]	SSRN [34]	CMR-CNN [35]	DPRnet [36]	MCVT [37]	SS-MCNet
Class 1	100.00%	100.00%	100.00%	100.00%	89.91%	**100.00%**
Class 2	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
Class 3	99.22%	100.00%	100.00%	100.00%	100.00%	**100.00%**
Class 4	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
Class 5	47.98%	48.84%	43.15%	45.58%	43.22%	**50.12%**
Class 6	86.05%	90.70%	100.00%	84.27%	95.12%	**100.00%**
Class 7	100.00%	97.71%	100.00%	98.34%	100.00%	**99.10%**
Class 8	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
Class 9	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
Class 10	100.00%	100.00%	100.00%	97.87%	100.00%	**100.00%**
Class 11	75.57%	97.38%	91.80%	82.79%	89.02%	**97.80%**
Class 12	62.30%	64.10%	61.50%	63.80%	60.90%	**63.60%**
Class 13	98.97%	91.04%	77.97%	99.41%	89.57%	**100.00%**
Class 14	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
Class 15	86.25%	99.60%	92.03%	82.67%	98.80%	**100.00%**
Class 16	87.64%	86.55%	64.64%	78.09%	84.60%	**90.45%**
Class 17	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
Class 18	85.77%	82.09%	80.74%	92.79%	92.05%	**98.50%**
Class 19	97.90%	100.00%	100.00%	100.00%	100.00%	**100.00%**
Class 20	100.00%	100.00%	100.00%	100.00%	100.00%	**100.00%**
OA	92.35%	94.62%	93.28%	93.86%	95.01%	**97.56%**
AA	90.13%	92.59%	90.93%	91.52%	92.76%	**95.43%**
Kappa	91.62%	93.98%	92.53%	93.17%	94.37%	**97.21%**



(a) (b)

Fig. 17. Classification maps for the comparison experiment between the fusion data and original data. (a) Fusion data classification result and (b) original data classification result. The bottom legend indicates the corresponding land-cover types represented by specific label colors.

TABLE X
CLASSIFICATION ACCURACY COMPARISON BETWEEN FUSION DATA AND ORIGINAL DATA

Class / Metric	Fusion Data	Original Data
Willow	96.74%	96.40%
Reed	99.98%	99.86%
Poplar	98.32%	99.79%
Cogon grass	92.24%	96.40%
Miscanthus	92.88%	98.63%
Road	99.87%	88.33%
Yellow River	100.00%	100.00%
Mudflat	100.00%	100.00%
OA	97.98%	98.08%
AA	97.75%	97.42%
Kappa	97.14%	97.75%

a notably high accuracy of 99.87% in the Road category, indicating that the enhanced spatial details effectively improve the recognition of categories with complex spatial structures. In contrast, the original data slightly outperform in the Poplar

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category with an accuracy of 99.79%, possibly because the additional spatial information introduced in this case, while enhancing representation, also brings certain redundant interference, which may slightly affect the model's discriminative ability. Both datasets achieve 100% classification accuracy for the Yellow River and Mudflat categories, suggesting that these classes are highly distinguishable regardless of spatial resolution. In terms of overall metrics, the original data shows slightly higher OA and kappa coefficient, highlighting the critical role of spectral information in overall classification stability. On the other hand, the fusion data demonstrates better AA, indicating more balanced performance across different categories, likely benefiting from the improved spatial resolution.

Despite being trained for only 20 epochs, the classification performance of the fusion data approaches that of the original data trained for 200 epochs, fully demonstrating the superior cross-source adaptability and training efficiency of the proposed SS-MCNet model. This experiment further verifies the complementary nature of spectral and spatial information in hyperspectral data fusion, offering a new technical pathway for multisource hyperspectral image classification and promoting the application of fusion methods in complex scenarios. However, we also acknowledge that the setting of training epochs is largely empirical and lacks systematic theoretical and experimental analysis as support. This issue will be a key focus of our future work to further optimize training strategies and enhance model performance.

V. Conclusion

The proposed SS-MCNet algorithm aims to enhance feature representation capabilities in hyperspectral images. It integrates three modules that incorporate multiscale perception units and position-aware modeling components, effectively capturing both multiscale and global features. By introducing SS-MCAF, the three feature streams are jointly optimized under the constraints of CL and DL, which not only preserves the complementarity among features but also significantly strengthens overall discriminative power, thereby improving the classification performance. Moreover, this study not only focuses on the design and training of deep learning models but also covers the entire process of hyperspectral data acquisition and preprocessing, establishing a complete and efficient remote sensing classification research pipeline. In addition, it proposes effective technical solutions for multisource data fusion, reflecting the dual value of the method in theoretical innovation and engineering practice.

The following two points outline my future research directions. First, future research will focus on designing modules that better capture the collaborative representation of spatial and spectral features, along with developing more efficient fusion mechanisms to enhance the model's capability in complex remote sensing scenarios. In this regard, emerging sequence modeling architectures such as Mamba hold great promise. As a novel model that has demonstrated superior performance over Transformers in long-sequence modeling tasks, Mamba offers new opportunities for spatial–spectral interaction modeling in hyperspectral remote sensing and warrants further investigation and empirical evaluation. Second, although this study benefits from a self-constructed hyperspectral–panchromatic multisource dataset, which facilitates experiments on multisource data fusion, the depth and breadth of the current experiments remain limited. The intrinsic information within the data has yet to be fully exploited. Future work will aim to advance classification tasks involving multisource remote sensing data by exploring deeper interactions among different modalities and systematically evaluating the adaptability and generalization of fusion methods across various land-cover categories and scene types.

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